

Gate B — Definition of Done Walkthrough

Phase 2 QA checkpoint, gate 2 of 4

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Run 2-6 September 2026 (Days 15-16). Record compiled 8 September 2026 (Day 17).

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Purpose

Gate B is the second of four gates in the Phase 2 QA checkpoint. Where Gate A asked “*is the code that ships the code we think ships*”, Gate B asks “**does each delivered item meet the definition of done written for it**”.

Twelve items, each carrying a written verdict:

P2.1, P2.2, P2.2b, P2.3, P2.4/P2.4b, P2.5, P2.6a, P2.7, P2.8, P2.8b, P2.8c, P2.8d

Three rules governed the run:

1. **Amended criteria are checked as amended, not as originally written.** Two DoD amendments were on record — “clicking a node” became “selecting from the list” (Amendment 1), and the taxonomy-skills line became enrichment-sourced (Amendment 2). Both were honest rewrites of criteria that could not otherwise pass, made before the build rather than at the checkpoint.
2. **Verdicts are recorded against the running app, not from memory.**
3. **Nothing is fixed inside the checkpoint.** Defects are named and phase-assigned.

The app was started once and left up for the walkthrough, with the port confirmed before anything else — a zombie server on a different port serving stale modules had already cost the project a session.

Verdict summary

Item	Verdict	Date
P2.1 — uncertain-node confirmation UX	PASS	2 Sept
P2.2 — self-enrichment	PASS	2 Sept
P2.2b — enrichment comparison	PASS	2 Sept
P2.3 — 3D canvas	PASS	2 Sept
P2.4 / P2.4b — real-time search	PASS with defect	2 Sept
P2.5 — profile card	PASS	2 Sept
P2.6a — group selection and dimming	PASS	2 Sept
P2.7 — cluster summaries	PASS	2 Sept
P2.8 — grounded chat panel	PASS	3 Sept, after D-16/D-17
P2.8b — shared display-table loader	PASS 8/8	6 Sept, after D-20
P2.8c — session save and restore	PASS 8/8	6 Sept, after D-23
P2.8d — review queue	PASS 7/7	6 Sept

Gate B closed 12 of 12 on 6 September 2026 (Day 16).

Session 1 — 2 September, P2.1 through P2.7

Eight items cleared in one sitting. Two produced findings.

P2.3 — a verdict corrected mid-gate

P2.3 was initially recorded against a scroll-wheel zoom failure. The user corrected the observation: **zoom works, but only via the UI button**.

That changed the verdict. The criterion says the user can zoom, not that the scroll wheel is the mechanism. **P2.3 — PASS:** 430 nodes rendered, UMAP-positioned, coloured correctly with greys distinguishable at depth, and rotate, zoom and pan all operational.

D-10 was withdrawn and folded into **D-11 — canvas interaction quality**, severity low, assigned to Phase 4. Four items, one likely root cause:

1. Scroll-wheel doesn't drive zoom; the button does
2. Wheel events over the canvas scroll the page instead
3. Mode buttons reset the view rather than composing
4. Canvas isn't pinned in the middle panel

Items 1 and 2 are the same bug seen from two sides — the wheel event reaches the page rather than the scene. Item 3 is `dragmode` defaults tuned for 2D. Item 4 is layout. Noted at the time: item 2 will feel worse on Community Cloud than locally, but it is an annoyance rather than a gate.

P2.4 / P2.4b — the one genuine FAIL

DoD as amended: *a search box filters and highlights nodes by name, role, SOC group, and any enriched skill terms, updating live as the user types; a query with no matches returns a clear empty state rather than a blank canvas.*

Check	Result
P2.4-a — partial name filters live, no Enter	FAIL
P2.4-b — SOC group name matches on group, not title text	PASS
P2.4-c — nonsense query gives clear empty state	PASS
P2.4-d — non-matches fade rather than vanish	PASS

P2.4-a failed against the written criterion. Requiring Enter is not “*updating live as the user types*”.

D-12 logged, severity medium, assigned to Phase 2 before deployment. It remained open through the rest of Gate B and was the last Phase 2 defect closed (Day 17).

Session 2 — 3 September, P2.8

P2.8's verdict was blocked by three defects at the point Gate B first reached it, and two more emerged during remediation.

	Defect	Resolution
D-13	Chat panel crashed on submit — illegal widget-key write after instantiation	Fixed via callback pattern with <code>PENDING_KEY</code>
D-14	Canvas didn't highlight chat answers — highlight written after <code>build_figure</code> in the same pass	<code>st.rerun()</code> after writing <code>HIGHLIGHT_KEY</code> , with a stated terminating condition
D-15	About text had no overlay in <code>loader.py</code> and could not reach <code>build_payload</code>	Deferred to P2.9a with the self-node identity question
D-16	Sonnet 5 is adaptive-thinking-only at effort high	<code>MAX_TOKENS</code> raised to 8000; early return on <code>stop_reason=max_tokens</code>
D-17	Harness could not reach a passing state while D-16 was live	Closed by D-16's fix; harness reached 31/31

D-16 — the finding worth keeping

At `MAX_TOKENS=4000`, the entire budget went to internal reasoning: 4000/4000 thinking tokens, block types `['thinking']`, **no text block opened at all**.

This corrected a prior recorded project learning. Day 14 had attributed a similar failure to “*starving a large answer*” of output tokens. That attribution was wrong — it is thinking-token consumption. `max_tokens` is a hard ceiling on thinking **plus** text combined, and at a low ceiling the model spends the whole budget before it writes anything.

The remediation was not simply a larger ceiling:

- `answer_question` now returns immediately on `stop_reason=max_tokens` with `NOTE_TRUNCATED`, rather than retrying. Retrying made it strictly worse — it added more input at the same ceiling.
- `_call` was extended to return a three-tuple including a `shape` string carrying block types and thinking-token counts, so the failure is diagnosable rather than merely observable.

- Verified by re-running `tools/diag_maxtokens.py: stop_reason: end_turn, thinking_tokens=3800`, block types `['thinking', 'text']`.

An enter-to-submit change was requested during this session and scoped into D-13's fix rather than deferred — `on_change=_submit_question` moved to the text input, Ask button removed.

Two minor defects were logged without fixing, per the no-fix rule:

- **D-18** — coverage counts disagreeing across three captions
- **D-19** — stray “no one matches” line on a question not about the table

Both were closed on Day 17.

Session 3 — 6 September, P2.8b / P2.8c / P2.8d

The final three items. Two produced defects, one of them in an instrument.

P2.8b — PASS 8/8, after D-20

D-20 was a checker defect, not a product defect, and it presented as the opposite of the truth.

PowerShell 5.1's `Set-Content -Encoding UTF8` writes a byte-order mark. Three files authored that way were invisible to `check_single_loader.py`, which expects BOM-free UTF-8. The checker then **conflated parse failures with rule violations** and reported those three files as violating the single-loader rule.

Two things had to be fixed. The instrument had to distinguish “I could not read this” from “this breaks the rule” — a conflation that would have reported nothing at all had it gone the other way. And all file authoring moved to `.NET WriteAllText` with `UTF8Encoding($false)`, including commit messages written via temp files.

D-22 was logged as a docs defect from this item: the P2.8b rationale cites 26 `apply_corrections` tests; the actual figure is 9.

P2.8c — PASS 8/8, after D-23

D-23 — a nested `st.expander` crashed `_render_report`.

Streamlit forbids nesting expanders. The restore report put one inside another to show non-matching entries. It had **never fired**, because every restore performed to that point had matched 100% of entries — the branch that renders misses had never executed.

Normal usage would have hit it immediately: re-export LinkedIn, some people changed jobs, restore, crash. It would have been a user's *first real restore*.

Fixed mid-gate with `st.container(border=True)` and re-observed.

The drift branch — the two-key notes design argued for since P2.7, where notes key on name alone and reattach with a flag when the person's title has moved — was also confirmed as never having run in the UI before this gate.

P2.8d — PASS 7/7

Clean. Three live checks:

- **a.** Assign from the queue — entry leaves, counter decrements, node recolours
- **b.** Skip an entry — leaves the view, **counter does not move**, person stays grey and findable by search. A skip is “not now” about a sitting, not a verdict on a person.
- **c.** Pagination at 10 per page — page refills sensibly as entries leave rather than throwing the user back to page one

D-24 — logged, then withdrawn

An apparent off-by-one between corrections restored and the counter:

Corrections restored	168 – n	Counter showed	Gap
38	130	131	1
42	126	128	2

The second data point produced the explanation. **A correction applied to an already-classified node does not reduce the needs-review count** — it changes a group, it does not resolve a grey node. The gap is simply how many corrections were made from a profile card rather than from the queue, and it should grow as more classified people are corrected. The `NOT_OCCUPATION` count moving 2 → 4 across the same interval is consistent with the same thing.

Closed as **not-a-defect** via `tools/reconcile_counter.py`, which proved the counter correct throughout.

Worth noting for the record: this finding reproduced independently on Day 17 against the 442-row table — 183 uncertain at rebuild, 148 after restore, 35 resolved against 37 corrections applied, a gap of 2. A better confirmation than the original script, because it was not looking for one.

What this gate bought

Three defects that no unit test could have reached, **all the same shape: a branch that had never executed**.

- **D-23** — the restore report crashed on any non-matching entry, behind a branch unreached because every prior restore was 100% clean.
- **D-20** — three files invisible to the single-loader checker, reported as violations, which is the opposite of the truth.
- **The drift branch** — argued for since P2.7, never once run in the UI.

In all three cases the code looked right and had been reasoned about carefully. What was missing was **an execution**. That is the entire justification for live harnesses alongside the unit suite, and this gate paid for itself.

Process finding

D-23 was fixed inside the checkpoint, which the no-fix rule forbids.

The fix was correct and the criterion was re-observed afterwards, so nothing in the verdict is unsound. It goes in the record as a **crossed line**, not as normal practice. The distinction matters: a gate that quietly absorbs its own fixes stops being evidence about the build it was run against.

Defect log at gate close (6 September)

	Defect	Status	Phase
D-11	Canvas interaction quality	open	4
D-12	Search requires Enter, no live filtering	open	2
D-18	Coverage means two things: summaries excludes small groups (289+8), chat includes them (297)	open	2
D-19	Stray “no one matches” on non-table questions	open	2
D-20	Checker conflated unparseable with non-compliant	fixed, uncommitted	2
D-21	Harnesses must be invoked <code>python -m tools.<name></code>	docs	—
D-22	P2.8b rationale cites 26 <code>apply_corrections</code> tests; actual is 9	docs	—
D-23	Nested expander crashed the restore report	fixed, uncommitted	2
D-24	Counter gap vs corrections restored	not-a-defect	—

All Phase 2 entries in this table were closed on Day 17 (8 September), together with D-25, D-26 and D-27 discovered during that work. D-11 remains open against Phase 4.

Note on this record

Compiled on Day 17 from the Day 15 and Day 16 session transcripts rather than from a running summary, so that the verdicts and the reasoning behind them are traceable to the observations that produced them.

Two areas are thinner in the transcript than the rest and are recorded here at the level of detail available: the individual criterion-by-criterion checks for P2.1, P2.2, P2.2b, P2.5, P2.6a and P2.7, which passed without producing findings and were recorded as clean; and the full ten-criterion breakdown for P2.8, which is represented above by its blocking defects rather than by each criterion’s verdict.